GEAR-Nav

Action-free spatial trajectory refinement via energy-based aggregation

Learn corrections from video.
Refine paths from vision.



Real-world robot deployments

Learn from video, without target paths



In-the-wild egocentric observation

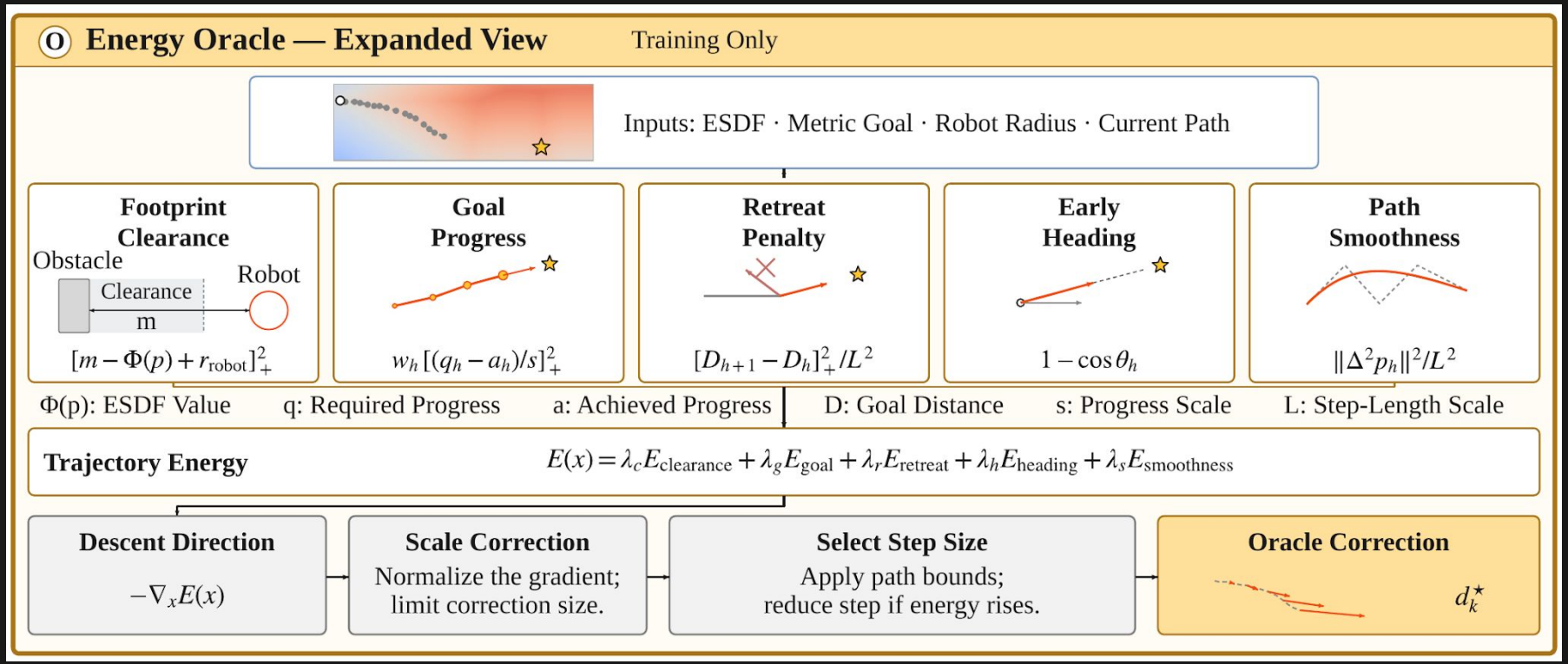
Video captures diverse navigation scenes, but provides no robot actions or target trajectories.

Use scene geometry to teach how a path should change.

Geometric energy supplies local correction targets during training.

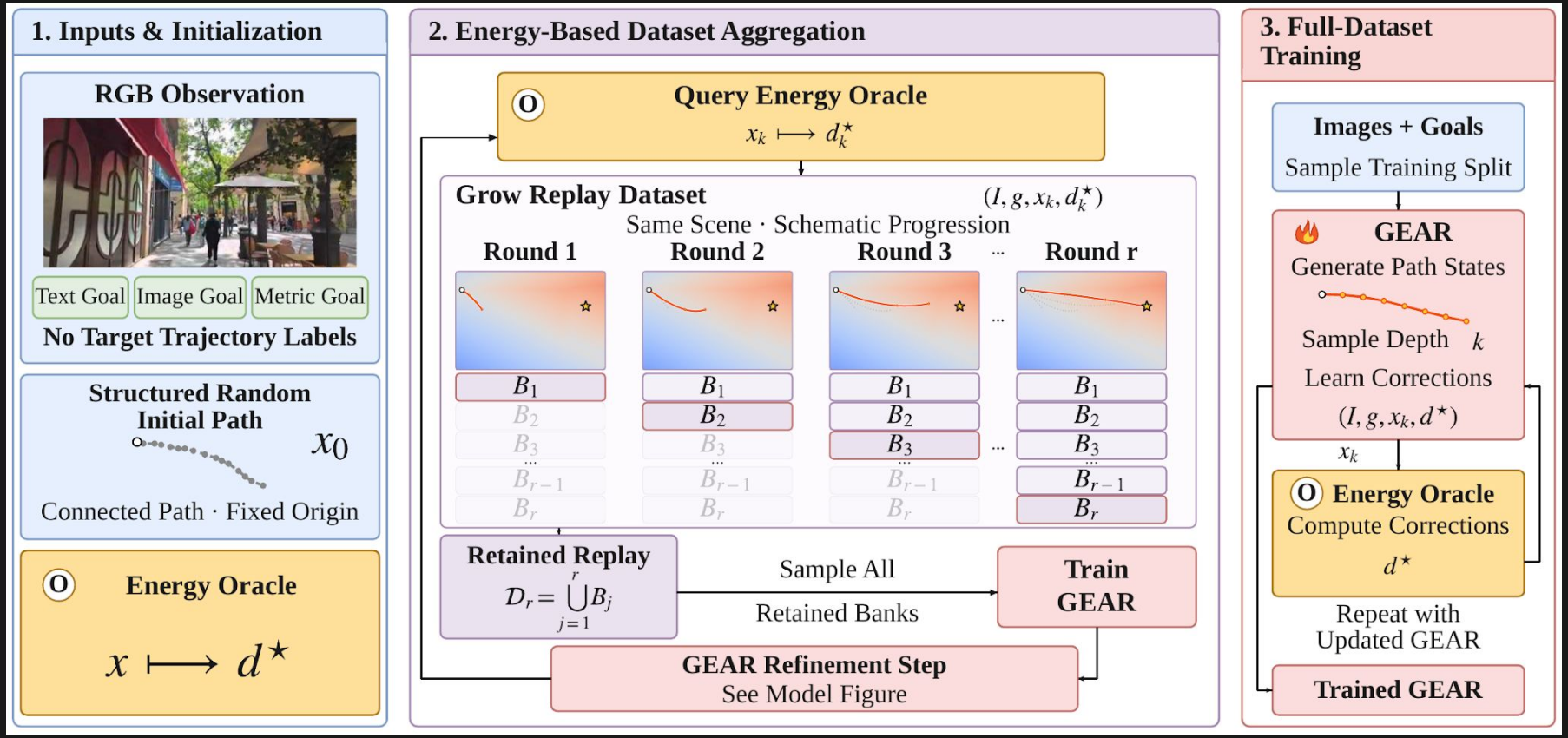
Geometry defines the training signal

Supervise local corrections with a projection-aware energy oracle.



Training geometry only; no optimized target trajectory labels.

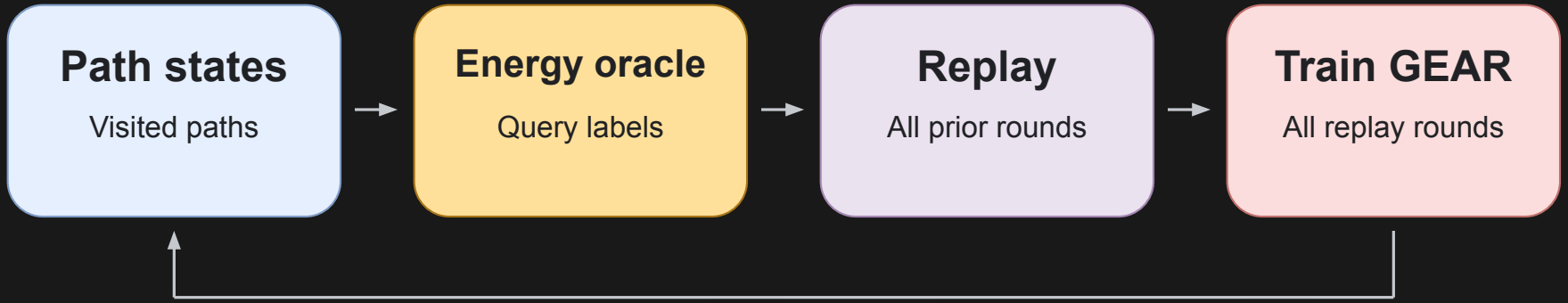
GEAR-Nav training pipeline



Oracle targets are aggregated across model-visited candidate paths.

Train on paths the model visits

Each learned update changes the next training state.

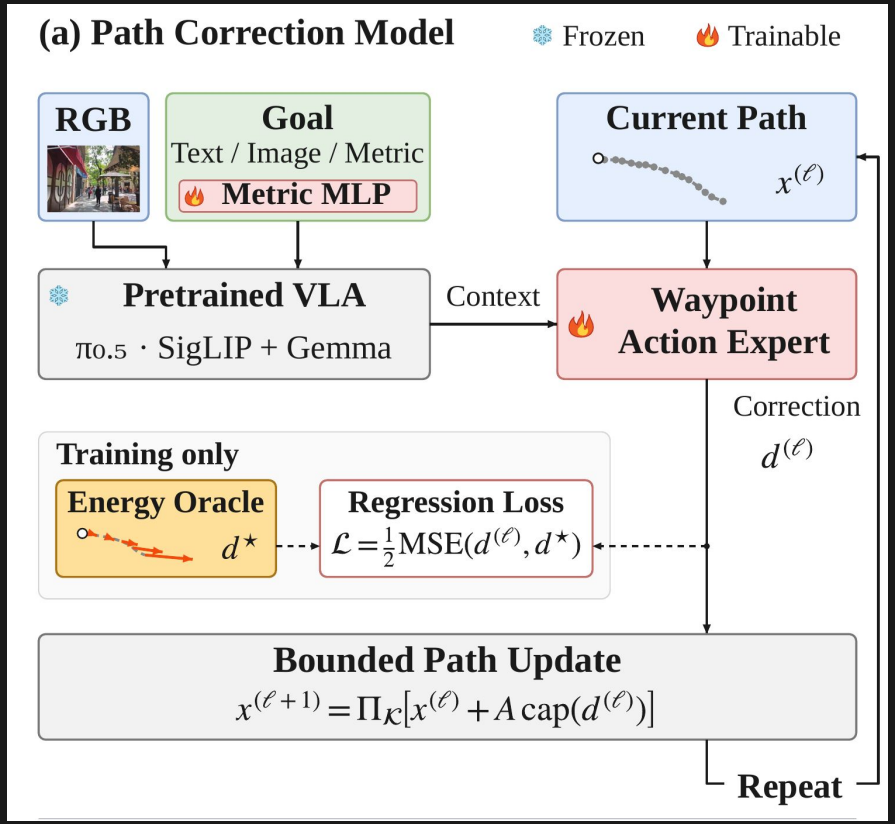


Advance the model-generated frontier

B_1 B_2 B_3 … B_r

Query corrections at visited paths; retain supervision from all earlier rounds.

GEAR model: learn a correction field



Condition on vision and the path

RGB and the navigation goal provide context; the current path enters the waypoint expert.

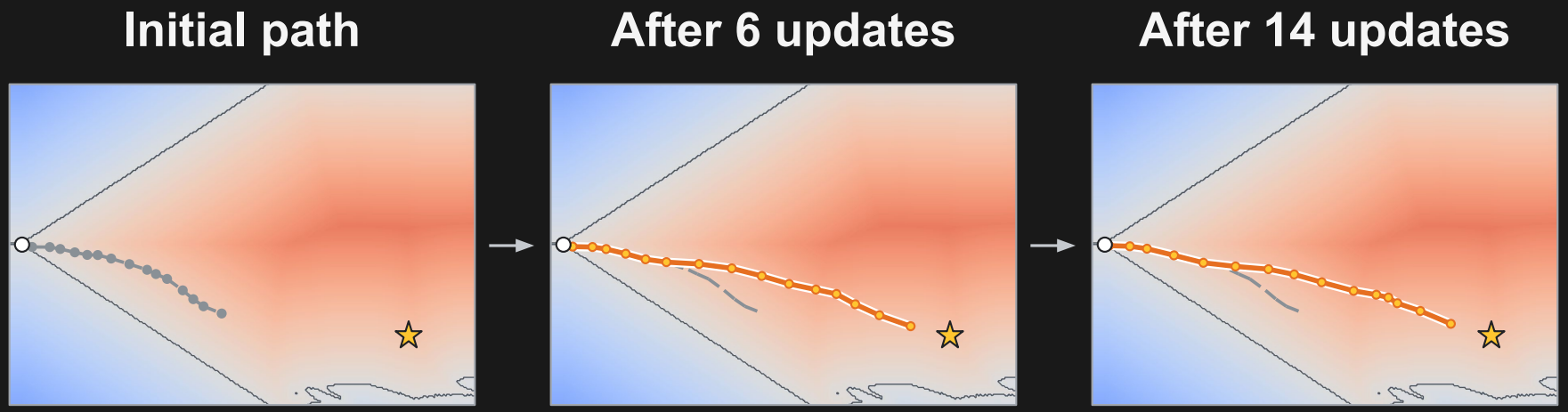
Predict a local correction

Learn the energy-oracle direction at each aggregated path state.

Apply a bounded update; repeat

At deployment, refinement uses no ESDF or analytic energy queries.

Recorded model-only refinement



Saved checkpoint example. ESDF is shown only as a geometric reference.

Around an intervening chair

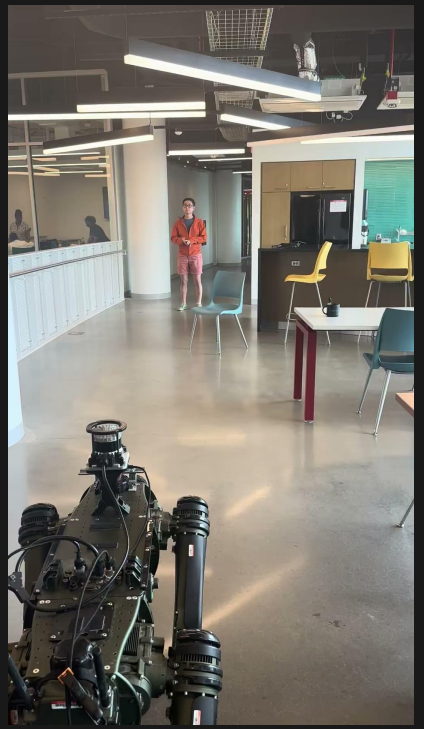
GEAR-Nav



OmniVLA



NaVILA



TIC-VLA



Goal: reach the person beyond the intervening chair

Separate trials · 1× playback · Baseline labels follow VEGA

Past a moving person

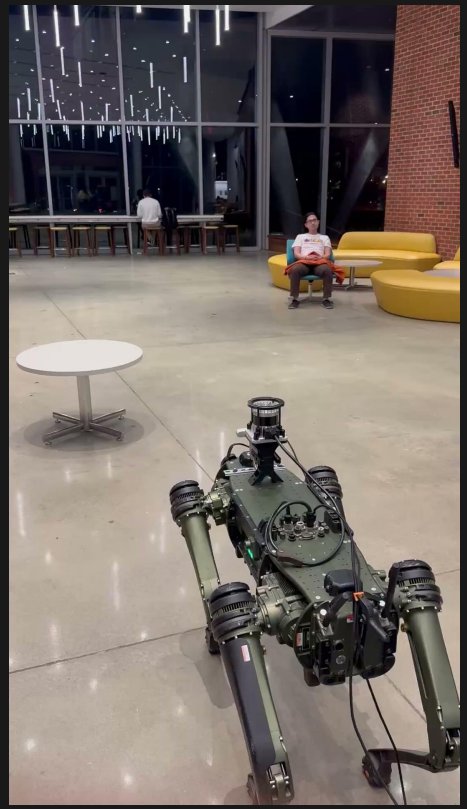
NAVIGATION GOAL

Blue chair in the lounge

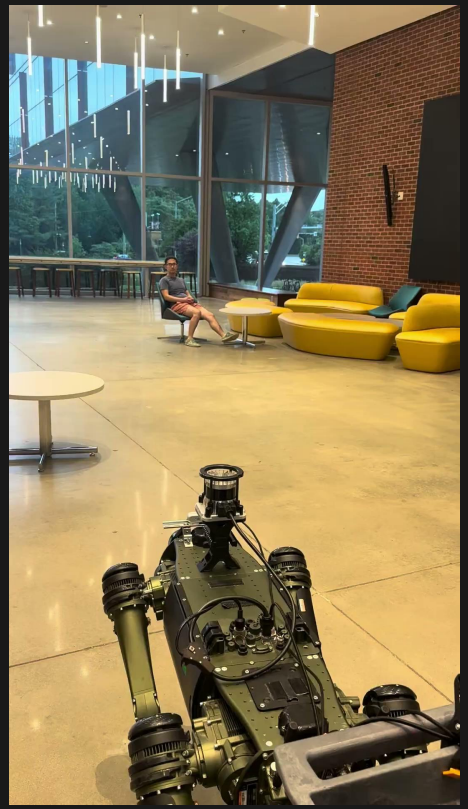
Reach the destination as a person crosses the route.

Representative separate trials · 1×

GEAR-Nav



OmniVLA



Scooter goal among pedestrians

GEAR-Nav



OmniVLA



NaVILA



Goal: scooter beyond the pedestrians

Separate trials · 1× playback

Additional robot deployments

Indoor



1× playback

Lounge



1× playback

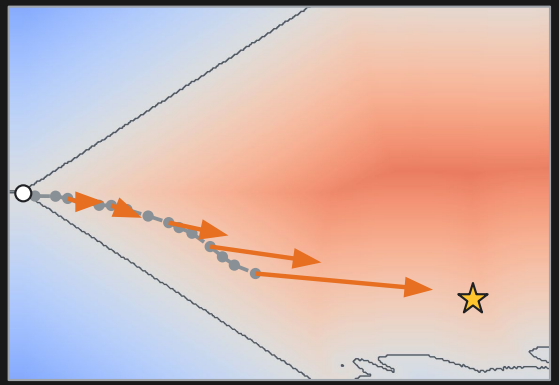
Outdoor



1× playback

From geometric supervision to visual refinement

TRAINING



Geometric supervision

Query local corrections

GEAR-Nav

Correction field

Aggregate labels at model-visited paths

DEPLOYMENT



Visual refinement

RGB + goal + candidate path

Learn corrections from geometry. Apply the learned field from vision.

GEAR-Nav

Action-free spatial trajectory refinement via energy-based aggregation

Geometric supervision during training.
Vision-conditioned refinement at deployment.